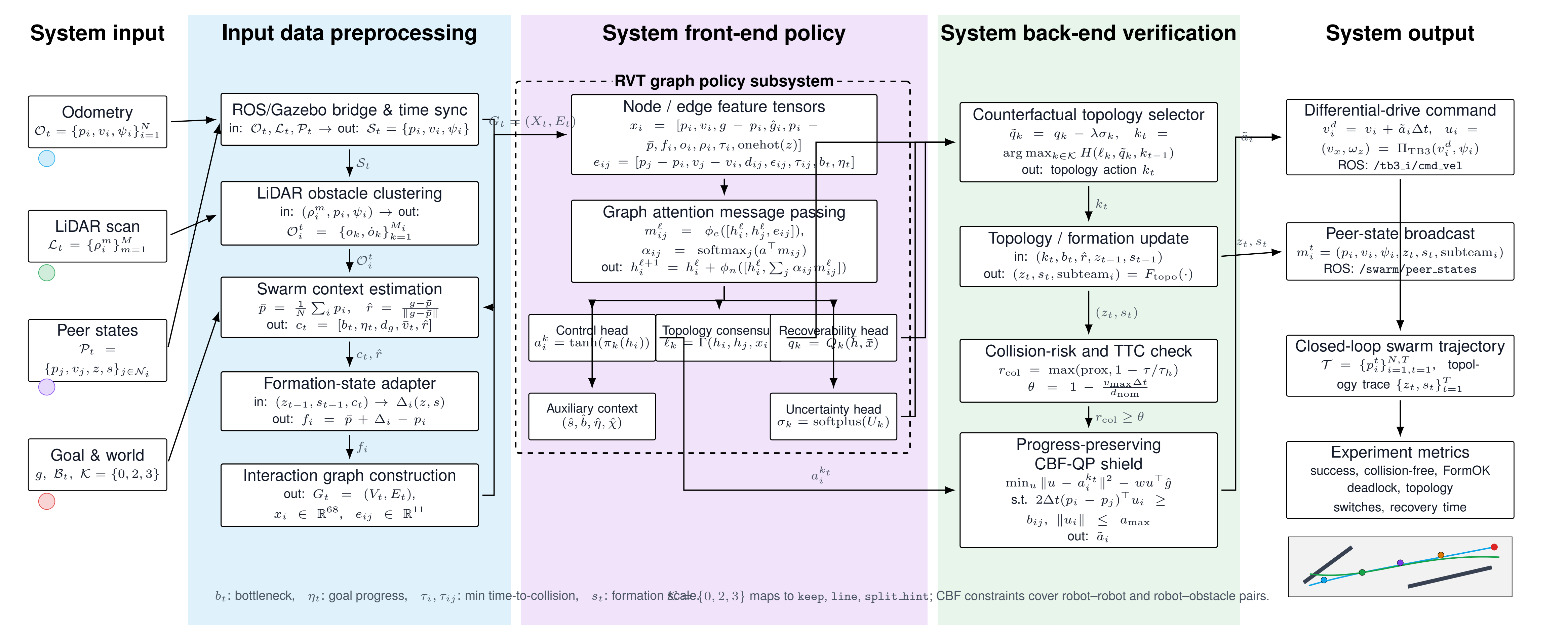

System input
Input data preprocessing
System front-end policy
System back-end verification
System output
RVT graph policy subsystem
Odometry
$\mathcal{O}_t = \{p_i, v_i, \psi_i\}_{i=1}^N$
LiDAR scan
$\mathcal{L}_t = \{\rho_i^m\}_{m=1}^M$
Peer states
$\mathcal{P}_t = \{p_j, v_j, z, s\}_{j\in\mathcal{N}_i}$
Goal & world
$g,\ \mathcal{B}_t,\ \mathcal{K} = \{0, 2, 3\}$
ROS/Gazebo bridge & time sync
in: $\mathcal{O}_t, \mathcal{L}_t, \mathcal{P}_t$ → out: $\mathcal{S}_t = \{p_i, v_i, \psi_i\}$
$\mathcal{S}_t$
LiDAR obstacle clustering
in: (ρ_i^m, p_i, ψ_i) → out: $\mathcal{O}_i^t = \{o_k, \dot{o}_k\}_{k=1}^{M_i}$
$\mathcal{O}_i^t$
Swarm context estimation
$\bar{p} = \frac{1}{N}\sum_i p_i,\quad \hat{r} = \frac{g-\bar{p}}{\|g-\bar{p}\|}$
out: $c_t = [b_t, \eta_t, d_g, \bar{v}_t, \hat{r}]$
$c_t, \hat{r}$
Formation-state adapter
in: (z_{t-1}, s_{t-1}, c_t) → $\Delta_i(z, s)$
out: $f_i = \bar{p} + \Delta_i - p_i$
f_i
Interaction graph construction
out: $G_t = (V_t, E_t)$,
$x_i \in \mathbb{R}^{68},\ e_{ij} \in \mathbb{R}^{11}$
$G_t = (X_t, E_t)$
Node / edge feature tensors
$x_i = [p_i, v_i, g - p_i, \hat{g}_i, p_i - \bar{p}, f_i, o_i, \rho_i, \tau_i, \mathrm{onehot}(z)]$
$e_{ij} = [p_j - p_i, v_j - v_i, d_{ij}, \epsilon_{ij}, \tau_{ij}, b_t, \eta_t]$
Graph attention message passing
$m_{ij}^\ell = \phi_e([h_i^\ell, h_j^\ell, e_{ij}]),$
$\alpha_{ij} = \mathrm{softmax}_j(a^\top m_{ij})$
out: $h_i^{\ell+1} = h_i^\ell + \phi_n([h_i^\ell, \sum_j \alpha_{ij} m_{ij}^\ell])$
Control head
$a_i^k = \tanh(\pi_k(h_i))$
Topology consensu
$\ell_k = \Gamma(h_i, h_j, x_i$
Recoverability head
$q_k = Q_k(\bar{h}, \bar{x})$
Auxiliary context
$(\hat{s}, \hat{b}, \hat{\eta}, \hat{\chi})$
Uncertainty head
$\sigma_k = \mathrm{softplus}(U_k)$
Counterfactual topology selector
$\tilde{q}_k = q_k - \lambda\sigma_k,\quad k_t = \arg\max_{k\in\mathcal{K}} H(\ell_k, \tilde{q}_k, k_{t-1})$
out: topology action k_t
k_t
Topology / formation update
in: $(k_t, b_t, \hat{r}, z_{t-1}, s_{t-1})$
out: $(z_t, s_t, \mathrm{subteam}_i) = F_{\mathrm{topo}}(\cdot)$
(z_t, s_t)
Collision-risk and TTC check
$r_{\mathrm{col}} = \max(\mathrm{prox}, 1 - \tau/\tau_h)$
$\theta = 1 - \frac{v_{\max}\Delta t}{d_{\mathrm{nom}}}$
$r_{\mathrm{col}} \geq \theta$
$a_i^{k_t}$
Progress-preserving CBF-QP shield
$\min_u \|u - a_i^{k_t}\|^2 - w u^\top \hat{g}$
s.t. $2\Delta t(p_i - p_j)^\top u_i \geq b_{ij},\ \|u_i\| \leq a_{\max}$
out: $\tilde{a}_i$
$\tilde{a}_i$
Differential-drive command
$v_i^d = v_i + \tilde{a}_i\Delta t,\quad u_i = (v_x, \omega_z) = \Pi_{\mathrm{TB3}}(v_i^d, \psi_i)$
ROS: /tb3_i/cmd_vel
z_t, s_t
Peer-state broadcast
$m_i^t = (p_i, v_i, \psi_i, z_t, s_t, \mathrm{subteam}_i)$
ROS: /swarm/peer_states
Closed-loop swarm trajectory
$\mathcal{T} = \{p_i^t\}_{i=1,t=1}^{N,T}$, topology trace $\{z_t, s_t\}_{t=1}^T$
Experiment metrics
success, collision-free, FormOK
deadlock, topology
switches, recovery time
b_t: bottleneck, η_t: goal progress, τ_i, τ_{ij}: min time-to-collision, s_t: formation scale. $\{0, 2, 3\}$ maps to keep, line, split_hint; CBF constraints cover robot–robot and robot–obstacle pairs.